

Class Ophiuroidea

Brittle stars and basket-stars

Occurrence 2,166 spp. worldwide; on the seabed

Although they are often mistaken for starfish, brittle stars and basket-stars form a separate and distinctive group of echinoderms. They have 5 long, narrow arms which, unlike those of starfish, are sharply demarcated from the small, central disk. The disk measures up to 4 in (10 cm) across, while the entire body can be up to 20 in (50 cm) in diameter. Brittle stars are the most mobile of all

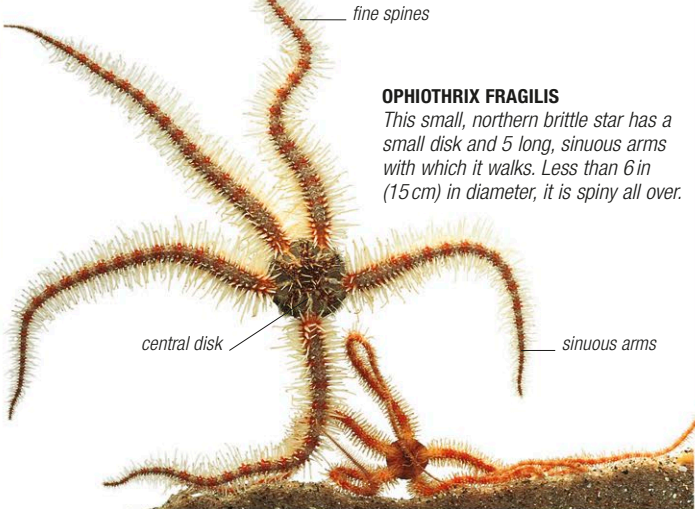
echinoderms and, like many of their relatives, can regenerate their arms if these are lost. Unlike starfish, brittle stars are either scavengers, feeding on dead animals, or suspension feeders, collecting small particles of food drifting down to the seabed, helped by a sticky mucus that is strung between the spines on their arms. Brittle stars are extremely abundant on some parts of the ocean floor, forming writhing masses that contain millions of animals. In contrast, basket-stars feed on their own, holding their intricately divided and branched arms at right angles to the current, so that they can catch small animals drifting past.



OPHIOTHRIX SPP.
These brittle stars may occur in dense aggregations on both rocky and sandy areas of the seabed. They have a diameter of up to 8 in (20 cm). Their spine-fringed arms project up into the water and filter feed.



ASTROBOA NUDA
Each arm of a basket-star branches repeatedly to give rise to a mass of armlets up to 30 in (75 cm) in diameter. In the Indo-Pacific species shown above, each armlet ends in tendrils. At night, the basket of arms is held up into the water to catch small swimming animals.



OPHIOTHRIX FRAGILIS
This small, northern brittle star has a small disk and 5 long, sinuous arms with which it walks. Less than 6 in (15 cm) in diameter, it is spiny all over.

Class Holothuroidea

Sea cucumbers

Occurrence 1,800 spp. worldwide; almost all buried in the seabed, or on the seabed surface



With their sausagelike shape and leathery texture, sea cucumbers look very different from other echinoderms, yet their internal anatomy is typical of this group, with organs present in sets of 5. They have 8–30 tube feet around the mouth, arranged in 5 rows; these are modified into finger- or bushlike feeding tentacles. Sea cucumbers vary from less than 1¼ in (3 cm) to 3¼ ft (1 m) in length. Some sea cucumbers discharge sticky, sometimes toxic, threads through their anus; these can entangle a would-be predator.



PSEUDOCOLOCHIRUS VIOLACEUS
This brightly colored, reef-dwelling sea apple cucumber from the Indo-Pacific has a ring of branched tentacles around its mouth. After feeding, each tentacle in turn is inserted into the mouth to be wiped clean. This animal is 4 in (10 cm) long.

Class Echinoidea

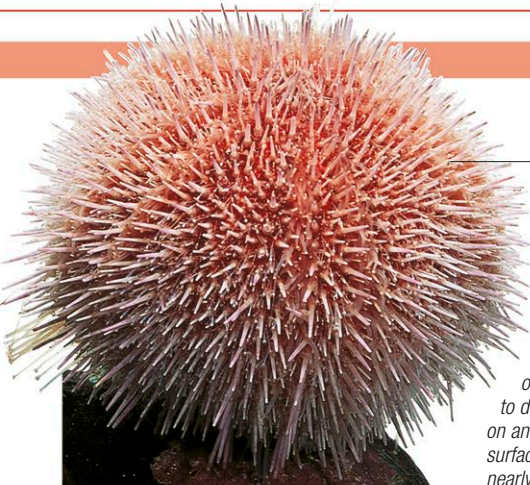
Sea urchins

Occurrence 1,090 spp. worldwide; on or buried beneath the seabed



Sea urchins typically have a spherical skeleton (test) of calcareous plates, covered by movable spines. Five rows of tube feet—often hidden by the spines—help these animals to creep over rocks, so that they can graze on algae and small animals. The mouth is on the underside, in the center, and

is armed with the “Aristotle’s lantern”—a set of 5 large, calcareous plates that work like jaws. After feeding, some sea urchins return to refuges that they have excavated over a period of time in soft rock; in the tropics, pencil urchins use their extra-thick spines to wedge themselves into crevices in coral reefs. The burrowing heart urchins and sand dollars are more disk shaped, and often have very short spines, in some species these look like backswept fur. Burrowing species are deposit feeders, mostly passing food to the mouth using the tube feet and spines. Sea urchins are usually 2¼–4¾ in (6–12 cm) in diameter, but some can grow to 14 in (35 cm).



spines and tube feet arranged in rows

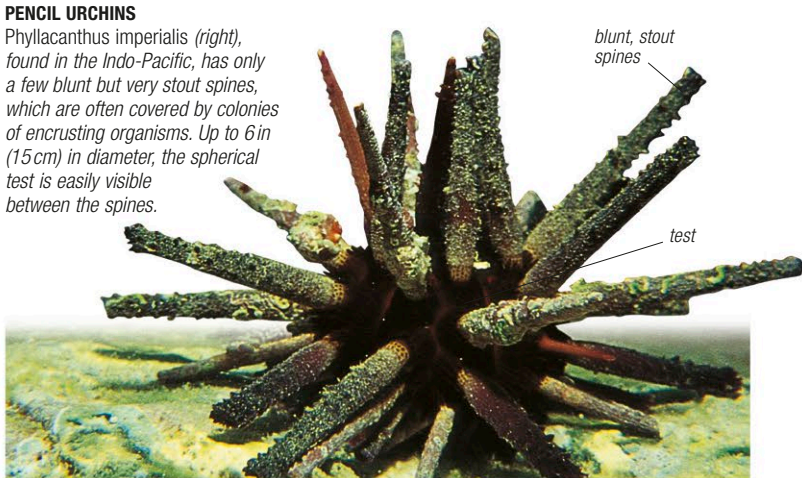
ECHINUS ESCULENTUS
The reproductive organs (gonads) of this large, edible sea urchin are prized as food around the shores of the North Atlantic. It occurs down to depths of 3,300 ft (1,000 m), grazing on animals and algae found on rocky surfaces. This species measures nearly 8 in (20 cm) in diameter.



DIADEMA SP.
These warm-water sea urchins, 12 in (30 cm) in diameter, have long, thin, brittle spines covered by glandular tissue that secretes a poisonous substance, which can cause severe irritation to human skin. Small fishes and crustaceans often live among the sea urchin’s spines, which offer them protection from predators.



SAND DOLLARS
Burrowing sea urchins such as this five-notched sand dollar lack the spherical test and long spines of other species. Like the tropical species pictured, Encope michelini, they have flattened tests, up to 6 in (15 cm) across, with a slight dome and short spines.



PENCIL URCHINS
Phyllacanthus imperialis (right), found in the Indo-Pacific, has only a few blunt but very stout spines, which are often covered by colonies of encrusting organisms. Up to 6 in (15 cm) in diameter, the spherical test is easily visible between the spines.

blunt, stout spines
test